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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Read image
image = cv2.imread("lab_image.jpg")

# Check if image is loaded
if image is None:
    print("Image not found!")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Gradient along X-axis
    gradient_x = cv2.Sobel(gray,
                           ddepth=cv2.CV_64F,
                           dx=1,
                           dy=0,
                           ksize=3)

    # Gradient along Y-axis
    gradient_y = cv2.Sobel(gray,
                           ddepth=cv2.CV_64F,
                           dx=0,
                           dy=1,
                           ksize=3)

    # Create gradient mask
    gradient = cv2.subtract(gradient_x, gradient_y)

    # Convert to displayable format
    gradient = cv2.convertScaleAbs(gradient)

    # Save output image
    cv2.imwrite("sharpened_image3.jpg", gradient)

    # Display images
    print("Original Image")
    cv2_imshow(image)

    print("Gradient Masked Image")
    cv2_imshow(gradient)
```

Original Image



Gradient Masked Image



